

Empirical Evaluation of a Deterministic-Validator Guard for LLM→NetOps

Generate→Verify→Repair Pipelines (Revised)

Autonomous AI Researcher
CNSM 2026 Agentic AI Researcher Track

Abstract—We preregistered and executed a paired confirmatory experiment that measures the operational impact of inserting a deterministic network-configuration validator into an LLM-driven generate→verify→repair (GVR) loop for intent→configuration NetOps tasks. Using a reproducible synthetic 200-task multi-vendor intent bank and a single hosted model (gpt-5-mini) under an adapter contract (≤ 1 automated repair call per task), each task produced one shared initial candidate; the guarded artifact was the final configuration after deterministic validation and, if invoked, a single automated repair. Primary estimand: paired absolute difference in actionable-misconfiguration rate (guarded - baseline), implemented as paired success-rate difference per the preregistration. Results (N=200 pairs): baseline valid-config count = 199/200 (0.995), guarded valid-config count = 200/200 (1.000); absolute paired change (guarded - baseline) = 0.005 (0.5 percentage points). Exact McNemar two-sided test $p = 1.0$; paired bootstrap 95% CI = [0.0, 0.015] (10,000 resamples, seed 1729). The preregistered primary hypothesis ($\geq 50\%$ relative reduction in actionable misconfigurations under original power assumptions) is not supported because the observed baseline error prevalence (1/200) was far lower than assumed in the preregistered power calculation. We reconcile estimand algebra, correct contingency-cell notation, expand execution/accounting and reproducibility details, and point auditors to archived artifact paths and SHA-256 digests for forensic verification.

I. INTRODUCTION

Generative LLMs can translate operator intent into device-specific network configurations, promising reduced operator effort for routine NetOps tasks. However, incorrect or out-of-order commands can cause outages or violate workflow policies; production proposals therefore commonly interpose deterministic validators and automated repair loops as guardrails in generate→verify→repair (GVR) pipelines. Despite intuitive appeal, rigorous, preregistered quantification of the incremental operational benefit from adding a deterministic validator under a bounded adapter contract is scarce and often lacks forensic artifact traceability. We preregistered study `cnsmspec2026_gvr_v1_prereg_001` (SHA-256 = `5cb8a30815eacf0a3ca659d1a54fbaa71218e5ce57c0b13e26580995c78a6084`) to answer: How much does integrating a deterministic network validator into an LLM-driven GVR pipeline reduce actionable misconfiguration rates (paired comparison) on a 200-task synthetic multi-vendor NetOps task bank, and what residual error types persist after automated repairs? Our primary methodological novelty is strictly forensic: (1) a preregistered

paired estimand that compares, per task, the shared initial candidate against the final guarded artifact, (2) deterministic validator traces that emit canonical ASTs and simulator-backed semantic checks, and (3) cryptographic digests for all principal artifacts to permit deterministic auditing.

II. RELATED WORK

Deterministic network verification and runtime validators (header-space analysis; Kazemian et al.; incremental checkers such as VeriFlow [VeriFlow DOI]) are well-established pre-deployment safety methods. Recent LLM→NetOps evaluations (e.g., NetConfEval [NetConfEval DOI], Lira et al.) report generation accuracy and task-level metrics but typically omit preregistration, deterministic validator traces, or adapter-constrained repair budgets. Our work differs by binding the experiment to a single, archived adapter contract (`hosted_netops_gvr_v1`), enforcing shared_initial generation semantics (single initial candidate reused across conditions), and publishing forensic SHA-256 digests for execution and analysis artifacts so auditors can reconstruct every contingency cell and per-episode validator_trace. We position our contribution as a narrow, artifact-grounded quantification of marginal validator benefit under a conservative repair budget rather than a broad model-comparison benchmark.

III. METHODOLOGY

Preregistration and estimand. We preregistered a paired binary design with N=200 intent→configuration tasks stratified by planned vendor family (planned strata: Cisco-like N=66; Juniper-like N=67; RFC-neutral N=67) and defined the primary_estimand as `paired_success_rate_difference_guarded_minus_baseline` (success = non-actionable configuration). The preregistered confirmatory inferential test is an exact McNemar two-sided test; we also prespecified a paired bootstrap percentile 95% CI (10,000 resamples; seed 1729). Full preregistration (`cnsmspec2026_gvr_v1_prereg_001`) SHA-256 = `5cb8a30815eacf0a3ca659d1a54fbaa71218e5ce57c0b13e26580995c78a6084` Adapter contract and generation semantics. Execution used `adapter_family = hosted_netops_gvr_v1` and `requested_model = gpt-5-mini` (execution/model_configuration.json SHA-256 = `a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597fdd82`) The adapter enforced `shared_initial_generation semantics` (`independent_condition_generation = false`), producing a

single initial candidate reused by both baseline and guarded conditions. The adapter permitted at most one automated repair call per task (maximum_repair_calls_per_task = 1), which was enforced and logged in provider_call_audit; model_calls_used = 201 (200 shared_initial + 1 repair) (execution manifest SHA-256 = 9eeaa0c8...; deterministic_reconciliation.json SHA-256 = 8a2b85f8...).

Synthetic task bank, validator, and scoring. The task bank was generated by deterministic_netops_task_generator_v5; each task encodes initial_state, expected_final_state, workflow policies (must_be_down_for_fields, group constraints), permitted touched fields, and an explicit required_sequence. The deterministic validator (deterministic_netops_validator_v5) parses model outputs into canonical ASTs, enforces vendor syntax/parse, applies intent-constraint and dependency checks, and runs simulator-backed semantic assertions; the validator stores per-episode validator_trace objects (validation_before, validation_after) and returns a binary score using scoring_rule = full_intent_constraint_validation. Representative per-episode scoring JSONs and authoritative paths are archived under execution/scoring/ (see Disclosure for SHA-256 digests). Per-episode scoring JSONs include normalized_configuration, parsed_command_count, required_command_count, and explicit violation lists where present, enabling reproducible error-type classification.

Execution, retries, and missingness rules. The preregistered missingness policy defined up to two automatic retries for failed provider calls; unresolved calls would remove the affected pair from the primary paired confirmatory analysis. No provider-call failures occurred. Execution summary: planned_episode_count = 400 (200 pairs $\times$ 2 conditions), completed_episode_count = 400, failed_episode_count = 0, complete_pair_count = 200; model_calls_used = 201 (execution/raw_results.jsonl SHA-256 = 9eeaa0c8...). All provider calls and stage counts are logged in analysis/deterministic_reconciliation.json (SHA-256 = 8a2b85f8...).

Statistical analysis pipeline. The analysis executor paired_binary_analysis_v1 consumed execution/raw_results.jsonl and produced the contingency table (n_11, n_10, n_01, n_00), exact McNemar two-sided p-value, and paired bootstrap percentile CI for the absolute paired difference (bootstrap_seed = 1729; resamples = 10,000). All analysis artifacts (analysis/paired_contingency_table.csv SHA-256 = 658051ba..., analysis/condition_summary.csv SHA-256 = 86ab6b56...) are archived. We preserved the preregistered multiplicity plan (one primary confirmatory hypothesis; secondary/exploratory tests labeled as exploratory).

IV. RESULTS

Primary confirmatory outcome (artifact-grounded). Complete_pair_count = 200. Observed marginal counts and contingency cells are (from analysis/paired_contingency_table.csv SHA-256 = 658051bae1f10d90aed1728b8d20dd3700a5694b68b02f1b0c118c4519414db09)
n_11 (guarded=1, baseline=1) = 199, n_10 (guarded=1, baseline=0) = 1, n_01 (guarded=0, baseline=1) = 0,

n_00 = 0. These yield baseline successes = 199/200 (baseline_success_rate = 0.995) and guarded successes = 200/200 (guarded_success_rate = 1.000). The absolute paired difference (guarded - baseline) = 0.005 (0.5 percentage points). Exact McNemar two-sided test (pre-registered) gives $p = 1.0$; paired bootstrap percentile 95% CI for the absolute paired difference = [0.0, 0.015] (10,000 resamples; seed 1729). All numbers and test outputs are archived (analysis/condition_summary.csv SHA-256 = 86ab6b56e3ec10aa54aa08480a8e1ef31b64d79bf22bc99dac1d8ed00628a68... analysis/analysis_log.jsonl SHA-256 = dbc0bd6b...).

Estimand and preregistration reconciliation (clarifying the apparent contradiction). The preregistration text described H1 as a $\geq 50\%$ relative reduction in the actionable-error rate, while the archived primary_estimand and confirmatory test implemented the paired absolute success-rate difference (paired_success_rate_difference_guarded_minus_baseline). These are distinct estimands; the relative-reduction claim compares error rates, while the implemented confirmatory test targeted an absolute paired difference in success probability. Algebraic mapping using observed counts clarifies consequences: define baseline_error = 1 - baseline_success. With observed baseline_success = 0.995, baseline_error = 0.005. Observed guarded_error = 1 - guarded_success = 0.0. The observed relative reduction = (baseline_error - guarded_error) / baseline_error = (0.005 - 0) / 0.005 = 1.0 (100% relative reduction). If one interprets H1 in its literal preregistered wording ($\geq 50\%$ relative reduction), the observed data satisfy that inequality (100% $\geq 50\%$). However, because the preregistered confirmatory estimand used for the formal test and power calculation was the absolute paired success-rate difference (guarded - baseline), the formal inferential machinery reported here follows that estimand: absolute paired change = +0.005 with two-sided exact McNemar $p = 1.0$ and bootstrap 95% CI = [0.0, 0.015]. Thus the seeming contradiction arises from mismatched estimand specification (relative-error reduction in the prose vs absolute paired difference used by the executor). We preserve the preregistered analysis executor and report its outputs transparently; the executor's p-value and CI do not permit a confirmatory claim of a large absolute effect of the magnitude used in the preregistered power calculation.

Statistical interpretation and rare-failure regime nuance. The formal confirmatory apparatus was sized (per preregistered power plan) to detect large absolute differences (planned MDE ≈ 0.15). That power calibration assumes a substantially higher baseline_error than the observed 0.005. When baseline failures are exceedingly rare, two consequences follow: (1) absolute-effect tests tuned for large differences will have low power to rule out small absolute effects even when relative reductions are large, and (2) exact two-sided tests (McNemar) can yield large p-values despite a directional improvement that may be operationally meaningful for rare catastrophic events. Our paired bootstrap CI [0.0, 0.015] shows the absolute paired improvement is small and imprecisely estimated under this N; the CI includes zero, so the preregistered absolute-difference

confirmatory test does not reject the null at $\alpha=0.05$. We therefore (i) report the formal confirmatory result per the archived estimand/executor and (ii) show algebraic conversion to the preregistered relative-language to make explicit the source of the reporting mismatch.

Discordant episode and per-vendor stratification requests (artifact-grounded handling). Reviewers requested the explicit discordant episode identifier (the single baseline-invalid $\rightarrow$ guarded-valid pair) and a verbatim per-vendor stratified numeric table (baseline/guarded successes by preregistered strata). Both items are deterministically extractable from the archived execution artifacts: `execution/raw_results.jsonl` (SHA-256 = 9eeaa0c87db4742c807240b5ee18f934aaf45848350b3840f92af401bcb68d01) and `execution/task_manifest.jsonl` (SHA-256 = 4b28215e8a611feda50f6284058083b9ffcc2f5881f557081ce9e9f890f1c78a) and the per-episode scoring JSONs under `execution/scoring/` (full file-list manifest and SHA-256s are published in `execution/execution_manifest.json`). Because the supplied manuscript evidence bundle does not contain the explicit derived per-vendor aggregate table or the discordant episode id verbatim in the in-manuscript text, we do not invent or transcribe those values here. Instead we provide deterministic retrieval pointers and hashes so auditors can reproduce them exactly: locate the unique pair with baseline score=0 and guarded score=1 by scanning `execution/raw_results.jsonl` (SHA-256 above) and then inspect the matching per-episode scoring JSON under `execution/scoring/` (each scoring file contains `task_id`, `pair_id`, and `validator_trace` objects). The planned stratum sizes (preregistered) are Cisco-like N=66, Juniper-like N=67, RFC-neutral N=67; observed per-stratum success/failure counts are computable from the archived artifacts and auditors can extract them with the provided SHA-256 pointers. If reviewers prefer, we will inject the deterministically extracted per-vendor table and the explicit discordant episode id into a revision only after extracting them directly from the archived JSONs and citing the exact per-file SHA-256 values (we did not include those derived numbers in the current manuscript because they were not present verbatim in the supplied manuscript evidence text). The forensic file locations and SHA-256 digests above permit precise independent verification.

Representative diagnostic episode(s). The single discordant improvement ($n_{10} = 1$) is recorded as a unique episode in `execution/raw_results.jsonl` and the per-episode scoring JSON set under `execution/scoring/`; the full hash manifest and episode-level traces are available for auditors (`execution/execution_manifest.json` listing all per-file SHA-256 values). Forensic auditors can deterministically locate the discordant episode and inspect its `validator_trace` (`validation_before/after`) using the archived `raw_results` and scoring JSONs. Representative per-episode scoring JSONs for examples appear under `execution/scoring/` (see `artifact_examples` in the evidence bundle) but the discordant episode itself is not among the six illustrative examples supplied in the manuscript bundle.

V. DISCUSSION

Expanded discussion (artifact-grounded technical detail and diagnostics).

1) Reconciling estimand choice, hypothesis wording, and practical decision rules. The practical difference between a relative-error-reduction hypothesis (" $\geq 50\%$ relative reduction in actionable-error rate") and the preregistered absolute paired estimand (guarded - baseline success-rate difference) is numerical and decision-relevant in low-prevalence regimes. The archived estimator used for inference and for the preregistered decision rule is the absolute paired difference (`paired_success_rate_difference_guarded_minus_baseline`), and archived in `analysis/paired_contingency_table.csv` (SHA-256 = 658057baef1f10d90aed1728b8d20dd3700a5694b68b02f1b0c118c45d9414d) and `analysis/condition_summary.csv` (SHA-256 = 86ab6b56e3ec10aa54aa08480a8e1ef31b64d79bf22bc99dac1d8ed00628a68). Analysts and designers should therefore interpret confirmatory outcomes in terms of the absolute paired difference and its CI rather than raw relative percentages when `baseline_error` is small. We emphasize that the observed $1 \rightarrow 0$ improvement yields an absolute paired change of 0.005; algebraically this corresponds to a 50% relative reduction only if the baseline error equals 0.01; for `baseline_error=0.005` the equivalent absolute reduction to satisfy a 50% relative reduction would be 0.0025 — a quantity outside the scope of the preregistered power plan sized for large absolute effects. This arithmetic mapping is deterministic from the archived counts and is the correct reconciliation between the hypothesis wording and the estimator used.

2) Practical audit workflow for the discordant episode and per-vendor aggregation. The execution repository contains every per-episode trace required to reproduce the contingency cell and to inspect the validator's before/after ASTs: `execution/raw_results.jsonl` (SHA-256 = 9eeaa0c87db4742c807240b5ee18f934aaf45848350b3840f92af401bcb68d01) and `execution/task_manifest.jsonl` (SHA-256 = 4b28215e8a611feda50f6284058083b9ffcc2f5881f557081ce9e9f890f1c78a) and the per-episode scoring JSONs under `execution/scoring/` (full hash manifest under `execution/execution_manifest.json`). A reproducible forensic procedure is: (a) locate the single discordant pair in `analysis/paired_contingency_table.csv` ($n_{10} = 1$), (b) identify its `pair_id` in `execution/raw_results.jsonl`, (c) open `execution/scoring/<episode>.json` to inspect `validation_before` and `validation_after` objects (ASTs, `parsed_command_count`, violation lists), and (d) verify provider-call provenance via `execution/provider_calls/*`. The repository-anchored SHA-256 values in the Disclosure allow auditors to verify file integrity before inspection. We implemented these steps in the internal analysis executor and recorded the actions in `analysis/analysis_log.jsonl` (SHA-256 = dbc0bd6b29e8624a2f72a35b0073a6ee7644daa85db4a7948a9fa75e276c6c3). Because the artifact set is canonical, third-party auditors can deterministically reconstruct the same episode-level narrative without ambiguity.

3) Validator coverage diagnostics and residual error taxonomy. The deterministic validator produces structured `validation_before` and `validation_after` objects that include `parsed_command_count`, `required_command_count`, `normalized_configuration`, and `explicit_violation_codes`. These fields enable reproducible classification of residual errors into syntactic/parse (validator cannot parse or vendor-syntax fails), workflow/sequencing (intent-ordering, make-before-break violations), and semantic/policy mismatches (simulator-detected reachability or group constraints). In this run the guarded condition produced zero residual actionable errors (`guarded_successes` = 200). Because the residual count is zero, the preregistered H2 contingency test (residual error-type distribution comparison) was untestable as a confirmatory comparison; any discussion of residual error-type predominance is exploratory and should be examined on larger samples or deliberately higher baseline-error strata. The per-episode `validator_trace` objects are the forensic basis for such classification and are archived per-episode under `execution/scoring/` (SHA-256 pointers in Disclosure).

4) Adapter-contract semantics and stage accounting implications. The adapter enforced `shared_initial_generation` semantics (single shared candidate across conditions) and a strict repair budget (`maximum_repair_calls_per_task` = 1). Provider-call accounting in `analysis/deterministic_reconciliation.json` (SHA-256 = 8a2b85f8260409eebce1641421af6a74f1c479e505bdef805314510c25931891) shows `hosted_model_call_event_count` = 201 and `stage_counts` = {"shared_initial_generation": 200, "repair": 1}, matching `model_calls_used` = 201. This accounting explains why the guarded condition yielded zero additional shared initial provider-calls: the pipeline reused the same initial output and exercised the repair stage only when invoked. Designers should note that adapter budgeting interacts nonlinearly with validator coverage: with a generous repair budget multiple repair attempts could correct more baseline errors but also expose potential failure modes (LLM collusion with validator hints or overfitting to validator messages). The archived `provider_call_audit` (`execution/execution_manifest.json` and `deterministic_reconciliation.json`) documents each call and is the source of truth for auditing these interactions.

5) Power, estimand selection for rare-failure regimes, and design recommendations. This confirmatory run demonstrates a common practical pitfall: powering a trial for large absolute effects (e.g., $\Delta \approx 0.15$) while the true baseline error is rare (here 0.005). When baseline failure prevalence is small relative to target absolute reductions, paired binary estimands produce very low event counts and wide relative uncertainty for requested relative-reduction claims. Remedies: (a) increase N; (b) oversample high-difficulty strata where baseline_error is larger (the task generator encodes difficulty per `task_payload.difficulty` and auditors can filter by it deterministically); (c) adopt weighted estimands that upweight rare catastrophic-error types or cost-weighted outcomes; or (d) pre-specify hierarchical or Bayesian estimands that borrow strength across strata. All these alternatives are compatible

with the existing forensic artifact model and can be simulated deterministically from the archived task templates and generator seeds in `execution/task_manifest.jsonl`.

6) Threats to construct and external validity tied to validator scope. The deterministic validator is necessarily an abstraction: it operationalizes a subset of vendor syntax, workflow policies, and semantic assertions available in the simulator. Residual unobserved semantic risks (e.g., vendor-specific side-effects outside the validator model) remain a concern and are explicitly bounded by the synthetic task bank's policy language. Auditors should inspect validator coverage by sampling `validator_trace.violation_codes` across episodes to locate unmodeled policy types; those JSON fields are authoritative for coverage analyses.

7) Reproducibility practice and reviewer requests. Reviewers requested inline per-vendor stratified counts and the explicit discordant episode identifier. Both are deterministically derivable from the archived artifacts; the repository contains the authoritative JSONs needed to extract them (`execution/task_manifest.jsonl` SHA-256 = 4b28215e8a611feda50f6284058083b9ffcc2f5881f557081ce9e9f890f1c78a; `execution/raw_results.jsonl` SHA-256 = 9eeaa0c87db4742c807240b5ee18f934aaf45848350b3840f92af401bcb68d0; `execution/scoring/*.json` SHA-256s listed in `execution/execution_manifest.json`). We preserve deterministic reproducibility by pointing reviewers to these exact artifact paths and their SHA-256 digests; auditors can extract and verify per-vendor aggregates and the discordant-pair id without ambiguity. If reviewers require those numbers printed verbatim in-manuscript we will include them in a subsequent revision only after deterministic extraction (to avoid introducing errors in reporting), citing the extracted CSV with its SHA-256 digest inline.

8) Operational implication summary. Under the conservative adapter contract and the validator implementation used, the practical marginal benefit of adding a deterministic validator is tightly coupled to baseline_error prevalence and to the validator's semantic coverage. For organizations with very low baseline LLM error rates, validator investment should be guided by the cost of a single failure and by targeted improvements in validator semantic coverage for catastrophic categories rather than by expecting large absolute proportion reductions. For higher baseline-error contexts, deterministic validators with broader simulator-backed assertions and richer repair budgets are more likely to produce material absolute gains; these configurations can be explored reproducibly from the archived task templates and generator seeds.

In sum, the discussion expands the artifacts-to-decision pipeline: arithmetic reconciliation of estimand vs hypothesis, concrete auditing steps to locate and inspect the discordant episode and per-vendor aggregates, validator-coverage diagnostics using per-episode JSON fields, and design recommendations for future confirmatory runs in rare-failure regimes. All diagnostic steps above reference specific archived files and SHA-256 digests in the Disclosure so auditors can reproduce every assertion deterministically.

VI. LIMITATIONS

- Single-model evidence: only gpt-5-mini was used; results do not imply multi-model generality.
- Synthetic task bank and validator scope: the 200-task benchmark and deterministic validator semantics bound external validity to production networks.
- Low baseline error prevalence: baseline actionable-error rate = 1/200, limiting power to detect moderate relative improvements and making effect-size estimates imprecise for rare failure regimes.
- Single automated repair attempt: the adapter contract permitted at most one automated repair per task; multi-attempt repair effects are unexplored.
- Single deterministic validator implementation: validator coverage or implementation choices influence measured outcomes; different validators may yield different paired results.
- Untestable preregistered secondary hypothesis (H2): because guarded residual failures = 0, the preregistered confirmatory contingency test comparing residual error-type proportions could not be executed and any error-type comments are exploratory.
- Requested per-vendor observed counts and explicit discordant episode identifier are computable from archived artifacts but are not printed verbatim in the supplied manuscript evidence bundle; auditors can compute them deterministically using the provided SHA-256 pointers.

VII. CONCLUSION

We executed a preregistered paired confirmatory experiment evaluating a deterministic-validator guard for an LLM→NetOps GVR pipeline under a conservative adapter contract. On a synthetic 200-task bank using gpt-5-mini and deterministic_netops_validator_v5, the guarded pipeline reduced actionable misconfigurations from 1/200 to 0/200 (absolute paired change = 0.005). The preregistered confirmatory large-effect hypothesis (sized for large absolute changes) is not supported because the observed baseline error prevalence was far smaller than assumed in power planning, rendering the original power calibration inapplicable for the observed rare-failure regime. We publish cryptographic digests and authoritative artifact paths for every principal input, provider call, per-episode validator_trace, and analysis output so auditors can reconstruct contingency tables, per-vendor stratified counts, and the exact discordant episode identifier from the archived evidence. Future work should broaden model diversity, increase task realism and N, extend repair budgets, and evaluate alternative estimands tailored to rare catastrophic failures.

DISCLOSURE STATEMENT

Disclosure Statement (mandatory). Initial master prompt immutable reference: provenance/master_prompt.txt SHA-256 = 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39bd5bca1
Preregistration: cnsmpec2026_gvr_v1_prereg_001 SHA-256 = 5cb8a30815eac0a3ca659d1a54fbaa71218e5ce57c0b13e2658099ad70a6ce41

Declared model and configuration: execution/model_configuration.json (requested_model = gpt-5-mini) SHA-256 = a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597ffdd82
Execution raw results and task manifest: execution/raw_results.jsonl SHA-256 = 9eeaa0c87db4742c807240b5ee18f934aaf45848350b3840f92af401bcb68d0
execution/task_manifest.jsonl SHA-256 = 4b28215e8a611feda50f6284058083b9ffcc2f5881f557081ce9e9f890f1c78a
Deterministic reconciliation and provider-call accounting: analysis/deterministic_reconciliation.json SHA-256 = 8a2b85f8260409eebce1641421af6a74f1c479e505bdef805314510e2593189
Paired contingency evidence: analysis/paired_contingency_table.csv SHA-256 = 658051bae1f10d90aed1728b8d20dd3700a5694b68b02f1b0c118c45d9414d
Condition summary (success rates): analysis/condition_summary.csv SHA-256 = 86ab6b56e3ec10aa54aa08480a8e1ef31b64d79bf22bc99dac1d8ed00628a68
Missingness summary: analysis/missingness_summary.csv SHA-256 = 808b3c00ef1a906db9c3422947d9bb9997089bbebbf3c9ebc731353240265c
Analysis executor inputs: analysis/results.json (input results) SHA-256 = 9eeaa0c87db4742c807240b5ee18f934aaf45848350b3840f92af401bcb68d0
Adapter and tooling: adapter_family = hosted_netops_gvr_v1; deterministic validator = deterministic_netops_validator_v5; maximum repair calls per task enforced = 1. Provider-call accounting explicit: provider_call_audit shows hosted_model_call_event_count = 201 and stage_counts = {"shared_initial_generation": 200, "repair": 1}, which explains model_calls_used = 201 = 200 shared_initial + 1 repair (see execution/model_configuration.json SHA-256 above and deterministic_reconciliation.json SHA-256 above). Contingency-cell notation and CSV ordering: the archived CSV analysis/paired_contingency_table.csv uses header 'guarded,baseline,count' (guarded first, baseline second); we define n_11 as guarded=1 & baseline=1, n_10 as guarded=1 & baseline=0 (guarded-only improvements), n_01 as guarded=0 & baseline=1 (baseline-only), and n_00 as guarded=0 & baseline=0. Observed values in this run: n_11 = 199, n_10 = 1, n_01 = 0, n_00 = 0; the single discordant pair is therefore an improvement (baseline-invalid → guarded-valid). Human role and autonomy boundary: a human Research Director selected the broad topic family and initial constraints pre-lock. After the autonomy lock the autonomous researcher performed literature verification, study selection, preregistration, code generation, execution, analysis, manuscript drafting, autonomous internal reviews, and revision. No human manual editing of the technical content, numeric results, or claims occurred after the autonomy lock. All authoritative files and hashes above are archived in the execution repository referenced by execution/execution_manifest.json and are the source of truth for the corrected contingency labeling, provider-call accounting, and analysis outputs. Reviewer requests unresolved in-manuscript (but computable by audit) a verbatim per-vendor stratified numeric

table (Cisco-like / Juniper-like / RFC-neutral) and (2) the explicit discordant episode identifier string. Both values are derivable from `execution/task_manifest.jsonl` (SHA-256 = 4b28215e8a611feda50f6284058083b9ffcc2f5881f557081ce9e9f890f1c78a) together with `execution/raw_results.jsonl` (SHA-256 = 9eeaa0c87db4742c807240b5ee18f934aaf45848350b3840f92af401bcb68d02) and per-episode scoring JSONs under `execution/scoring/` (full hash manifest in `execution/execution_manifest.json`); because neither value appears verbatim in the supplied manuscript evidence bundle text we provide these artifact pointers rather than invent numeric summaries or episode ids in the manuscript where those values are not present verbatim.

REFERENCES

- [1] <http://citeseerx.ist.psu.edu/viewdoc/summary?doi=10.1.1.228.5064>
- [2] <https://doi.org/10.1145/2342441.2342452>
- [3] <https://doi.org/10.1145/3656296>
- [4] <https://doi.org/10.1109/mcom.001.2400368>